Tian quartic Jacobian arithmetic gate

Receipt written: 2026-10-20T07:05Z UTC. **Command:** `tian-quartic-km` from `native/Cargo.toml`. **Host:** shared Intel Xeon 6975P-C; L0 operation counts only.

The native evaluation-space Khuri–Makdisi implementation constructed Tian's published quartic over F523 and the published classes S,T. It checked T=1987S, rS=0, inverse, commutative and associative controls. The known scalar 1987 is an input to this certificate; the run is not a destination-side DLP solver. The raw receipt is `correctness/tian-quartic-km-v2.json`.

At M+0.5S+10I, one measured quartic doubling costs 13318526 base-field multiplication equivalents, 14.043× the mean of five complete source-rho controls. Computing the deliberately known scalar 1987S costs 212.418× that mean. The complete price grid is `results/tian-km-arithmetic-gate-price-grid.csv`.

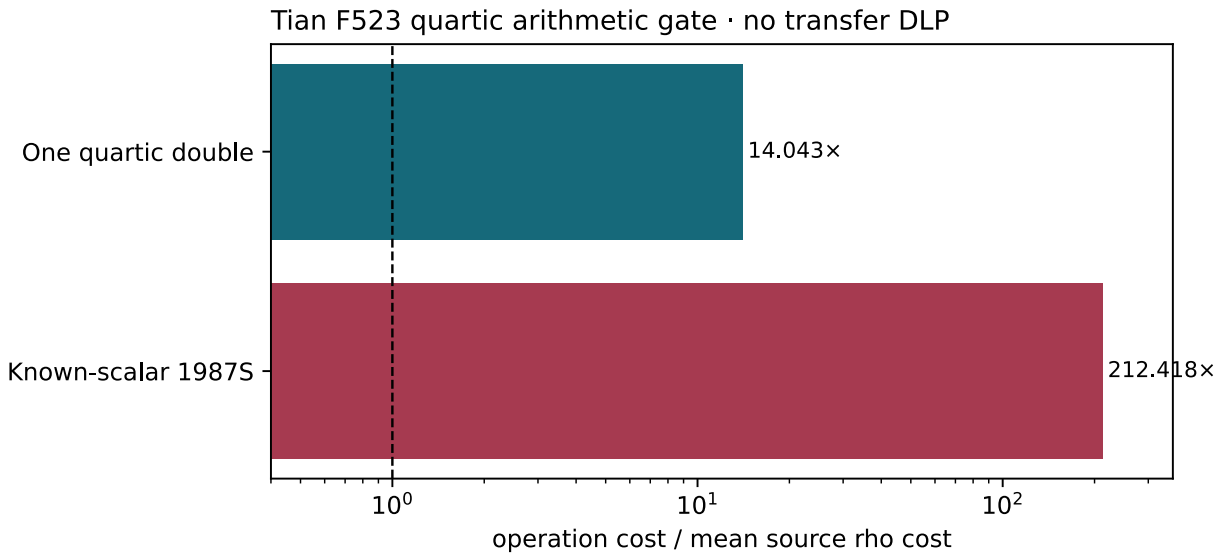

Figure 1: Recorded arithmetic gate ratios to the five-seed mean source-rho cost; no transfer DLP attack ratio is shown.

The source-to-Jacobian map, quotient kernel verification and a destination DLP solver remain unexecuted. No wall-speed or external-system comparison is claimed.